

Toxic Content Classification using BiLSTM

Problem

This project addresses the problem of classifying text into multiple safety- and crime-related categories. The data-set contains textual information derived from queries and image descriptions with a “Toxic Category” label. The goal is to build a LSTM model that capable of accurately classifying new text into one of nine predefined categories.

Dataset

Source: cellula toxic data (1).csv

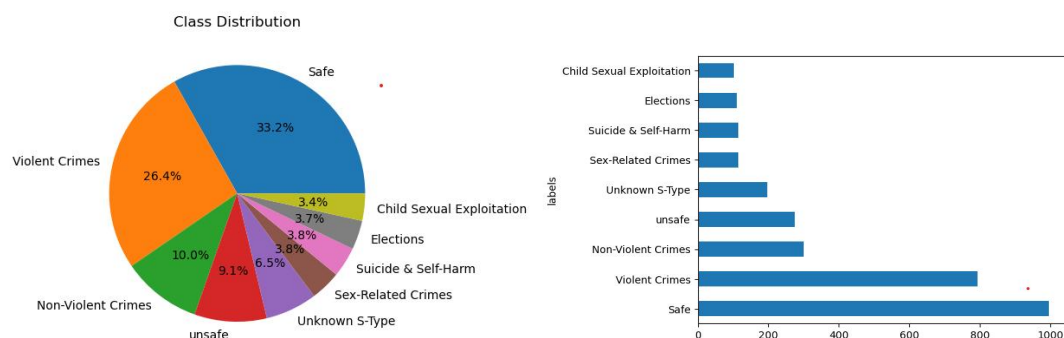
Features:

query and image descriptions were concatenated into a single text field.(X)

Label column: Toxic Category renamed to labels.(Y)

Classes: 9 (Safe, Violent Crimes, Elections, Suicide & Self-Harm, unsafe, Child Sexual Exploitation, Sex-Related Crimes, Non-Violent Crimes, Unknown S-Type).

Class Distribution: The 'Safe' class was the majority, representing a large portion of the data. Several toxic classes, such as 'Child Sexual Exploitation' and 'Unknown S-Type', had very few samples (fewer than 100). This imbalance posed a major risk for model bias.



Data Preprocessing

Text Cleaning

Lowercasing, removing digits, non-alphabetic characters, extra spaces.

Stopword removal using NLTK.

Splitting

The dataset was initially split into training and test subsets while preserving class balance.

A further split created a validation set for hyperparameter tuning and early stopping.

Augmentation

Minority classes were upsampled to at least 100 samples using two augmentation techniques:

Contextual Word Embedding Substitution (BERT).

Back Translation (EN→DE→EN using Facebook WMT19).

Augmented text was lightly re-cleaned.

Tokenization and Vocabulary

A simple whitespace tokenizer built a vocabulary with <pad> and <unk> tokens.

Texts were padded to a maximum length per dataset.

Model Architecture

Embedding Layer: `nn.Embedding` with `padding_idx` for <pad>.

Recurrent Layer: Bidirectional LSTM (`nn.LSTM`) with hidden size 128.

Dropout: 0.4 applied to concatenated hidden states.

Fully Connected Layer: Maps concatenated forward+backward hidden states to 9 output classes.

Loss: `CrossEntropyLoss` with computed class weights to mitigate imbalance.

Optimizer: Adam with weight decay regularization.

```
BiLSTMClassifier(  
    (embedding): Embedding(3302, 100, padding_idx=0)  
    (lstm): LSTM(100, 128, batch_first=True, bidirectional=True)  
    (fc): Linear(in_features=256, out_features=9, bias=True)  
    (dropout): Dropout(p=0.4, inplace=False)  
)
```

Training Procedure

Batch size: 32

Epochs: 20

Learning rate: 1e-3

L2 regularization : 1e-4

Early stopping: stops training when validation F1 does not improve for 5 epochs.

Metrics tracked: training/validation loss, accuracy, and weighted F1.

Experiments & Results

We conducted a comparative experiment by training two models:

Model A: Trained on the original, imbalanced training data

Model B: Trained on the augmented, training data.

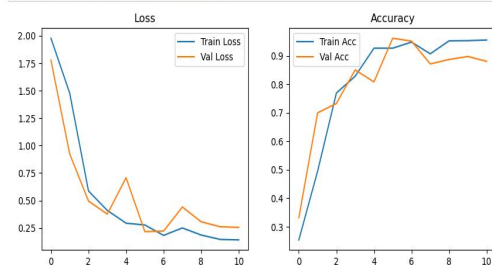
Both models were evaluated on the same held-out test set, which was never used in augmentation.

Model A

Epoch 1	Train Loss: 1.9780	Acc: 0.254	Val Loss: 1.7771	Acc: 0.332	F1: 0.220
Epoch 2	Train Loss: 1.4776	Acc: 0.495	Val Loss: 0.9241	Acc: 0.700	F1: 0.719
Epoch 3	Train Loss: 0.5867	Acc: 0.769	Val Loss: 0.4946	Acc: 0.733	F1: 0.798
Epoch 4	Train Loss: 0.4095	Acc: 0.828	Val Loss: 0.3750	Acc: 0.851	F1: 0.876
Epoch 5	Train Loss: 0.2933	Acc: 0.927	Val Loss: 0.2671	Acc: 0.889	F1: 0.890
Epoch 6	Train Loss: 0.2771	Acc: 0.927	Val Loss: 0.2166	Acc: 0.962	F1: 0.950
Epoch 7	Train Loss: 0.1822	Acc: 0.948	Val Loss: 0.2214	Acc: 0.952	F1: 0.948
Epoch 8	Train Loss: 0.2497	Acc: 0.907	Val Loss: 0.4400	Acc: 0.872	F1: 0.912
Epoch 9	Train Loss: 0.1854	Acc: 0.953	Val Loss: 0.3859	Acc: 0.887	F1: 0.919
Epoch 10	Train Loss: 0.1468	Acc: 0.953	Val Loss: 0.2613	Acc: 0.898	F1: 0.919
Epoch 11	Train Loss: 0.1410	Acc: 0.956	Val Loss: 0.2549	Acc: 0.881	F1: 0.908

Early stopping triggered.

gr(train_losses, val_losses, train_accs, val_accs)

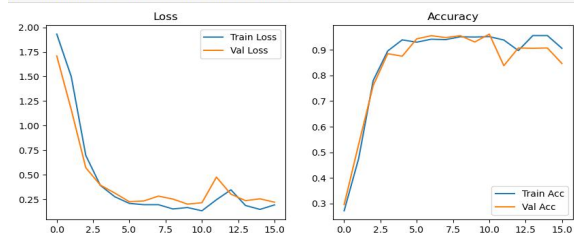


Model B

Epoch 1	Train Loss: 1.9320	Acc: 0.272	Val Loss: 1.7895	Acc: 0.296	F1: 0.264
Epoch 2	Train Loss: 1.4979	Acc: 0.476	Val Loss: 1.1617	Acc: 0.531	F1: 0.620
Epoch 3	Train Loss: 0.6960	Acc: 0.780	Val Loss: 0.5710	Acc: 0.761	F1: 0.761
Epoch 4	Train Loss: 0.3918	Acc: 0.896	Val Loss: 0.3946	Acc: 0.885	F1: 0.886
Epoch 5	Train Loss: 0.2752	Acc: 0.939	Val Loss: 0.3128	Acc: 0.875	F1: 0.877
Epoch 6	Train Loss: 0.2078	Acc: 0.930	Val Loss: 0.2249	Acc: 0.943	F1: 0.928
Epoch 7	Train Loss: 0.1947	Acc: 0.942	Val Loss: 0.2328	Acc: 0.956	F1: 0.938
Epoch 8	Train Loss: 0.1948	Acc: 0.940	Val Loss: 0.2019	Acc: 0.948	F1: 0.931
Epoch 9	Train Loss: 0.1511	Acc: 0.951	Val Loss: 0.2518	Acc: 0.956	F1: 0.941
Epoch 10	Train Loss: 0.1657	Acc: 0.950	Val Loss: 0.2006	Acc: 0.931	F1: 0.922
Epoch 11	Train Loss: 0.1326	Acc: 0.952	Val Loss: 0.2144	Acc: 0.962	F1: 0.947
Epoch 12	Train Loss: 0.2451	Acc: 0.939	Val Loss: 0.4751	Acc: 0.838	F1: 0.847
Epoch 13	Train Loss: 0.3460	Acc: 0.897	Val Loss: 0.3815	Acc: 0.908	F1: 0.908
Epoch 14	Train Loss: 0.1867	Acc: 0.956	Val Loss: 0.2361	Acc: 0.906	F1: 0.908
Epoch 15	Train Loss: 0.1466	Acc: 0.956	Val Loss: 0.2543	Acc: 0.908	F1: 0.898
Epoch 16	Train Loss: 0.1925	Acc: 0.906	Val Loss: 0.2280	Acc: 0.847	F1: 0.886

Early stopping triggered.

gr(train_losses0, val_losses0, train_accs0, val_accs0)

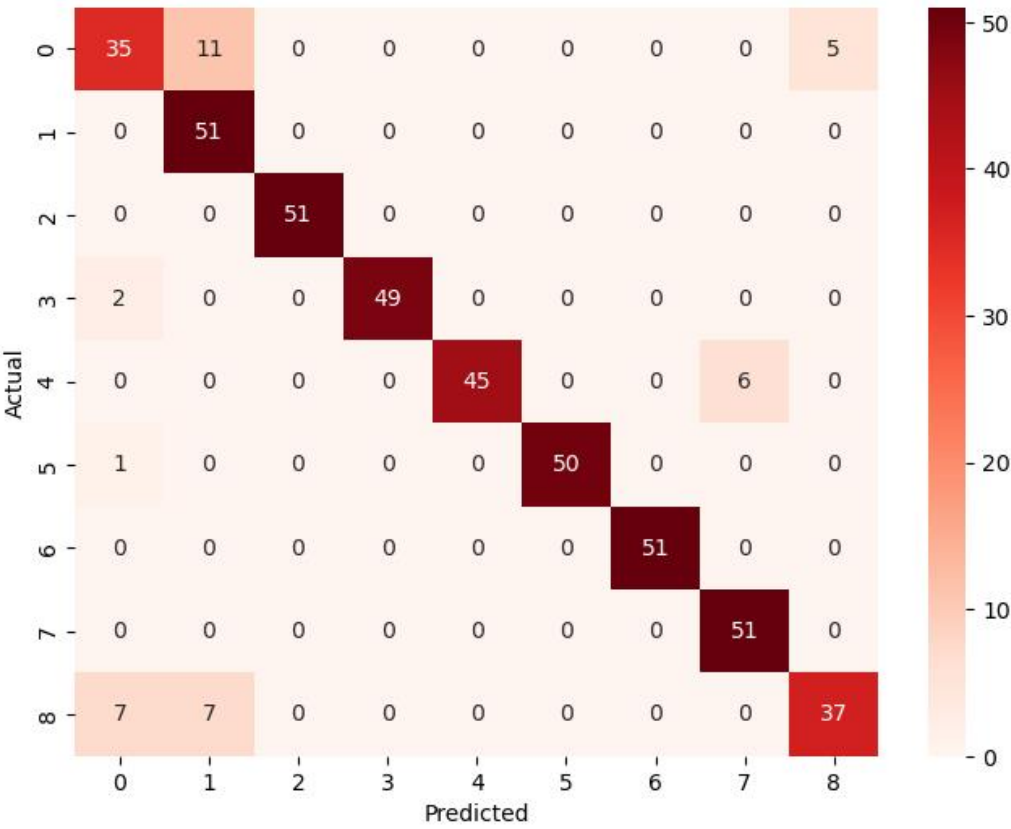


Results(on Test Set):

Metric	Model A (Imbalanced Data)	Model B (Augmented Data)
Weighted F1-Score	~91	~93
Weighted Precision	~92	~93
Weighted Recall	~91	~93

Model A

F1 Score: 0.9141
Precision: 0.9214
Recall: 0.9150



Model B

F1 Score: 0.9268
Precision: 0.9285
Recall: 0.9281

